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Ahmed Fouad Lagha

AI Engineer / Researcher

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Summary

PhD candidate in privacy-preserving ML with experience in deep learning, LLMs, and full-stack engineering. Track record of delivering research solutions, from hair segmentation models to production RAG pipelines. Seeking a Data Science or ML internship to solve real-world R&D challenges.

Technical Skills

Programming: Python, SQL, C/C++, Java, JavaScript, Bash
ML & Data Sci: PyTorch, Scikit-learn, OpenCV, Pandas, NumPy
Privacy ML: Federated Learning, Differential Privacy
Databases: Vector DBs, MySQL, SQLite, MongoDB
DevOps & Tools: Docker, Git, CI/CD

Experience

PhD Researcher — Privacy-Preserving ML — **Sep. 2025 – Present**
Eötvös Loránd University (ELTE) — *Budapest, Hungary*

- Designing DP-SGD federated learning pipelines for synthetic data ($\varepsilon = 1$) and benchmarking privacy-utility trade-offs
- Developing the open-source IDaSec Practicum (8 labs on ML security & privacy) and mentoring graduate students
- Awarded the Stipendium Hungaricum Scholarship for academic excellence and research potential

Software Engineer — **Dec. 2024 – Mar. 2025**
RayaneBio — *Remote*

- Architected and deployed a responsive e-commerce platform using React and Next.js, serving organic product customers
- Implemented secure user authentication and database models, improving user data management and transaction security
- Integrated customer feedback loops and testimonial features, elevating brand trust and increasing client engagement metrics

Projects

MyVellum — **Dec. 2025 – Present**
LLM Orchestration, RAG, Docker, CI/CD

- Architected an LLM orchestration system with modular prompt templates for an AI-powered autobiography platform
- Developed vector-backed RAG pipelines with context-stitching, reducing hallucination rates in generated responses
- Integrated automated evaluation gates into CI/CD for relevance and safety metrics, ensuring quality across deployments
- Owned end-to-end API design and Docker deployment; mentored two engineers on prompt engineering best practices

MS-AGHairNet — **Jan. 2025 – Jun. 2025**
Python, PyTorch, OpenCV

- Designed a deep learning architecture for hair segmentation, achieving a 95.37% Dice coefficient on the benchmark dataset
- Engineered a data augmentation pipeline that improved generalization across image conditions, reducing overfitting by 15%
- Evaluated model performance with precision, recall, and F1-score; optimized inference pipeline for real-time deployment

Education

PhD in Informatics — Privacy-Preserving Machine Learning — **Sep. 2025 – Expected 2029**
Eötvös Loránd University (ELTE) — *Budapest, Hungary*
Thesis: Synthetic data generation in artificial intelligence security (Supervised by Pr. Lendák Imre)

M.Sc. in Artificial Intelligence (GPA: 3.7/4.0) — **Sep. 2023 – Jun. 2025**
University of Khenchela — *Khenchela, Algeria*
Thesis: Hair Dermoscopic Image Segmentation Using Deep Learning (Supervised by Pr. Dalal Bardou)

B.Sc. in Mathematics and Computer Science (GPA: 3.9/4.0) — **Sep. 2020 – Jun. 2023**
University of Khenchela — *Khenchela, Algeria*

Certifications & Training

- **Software Engineering Specialization** — ALX Africa — Feb. 2025
- **AI Career Essentials (AiCE)** — ALX Africa — Jul. 2024

Languages

Arabic (Native) | English (Professional Proficiency) | French (Intermediate) | Hungarian (Beginner)